



Jothish Kumar P 251801123-2025-AIDS ▾

J2

Started on Thursday, 24 September 2026, 10:26 AM**State** Finished**Completed on** Thursday, 24 September 2026, 10:28 AM**Time taken** 1 min 50 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n , find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int findDuplicate(const int* arr, int n) {
5
6      int tortoise = arr[0];
7      int hare = arr[0];
8
9      do {
10         tortoise = arr[tortoise];
11         hare = arr[arr[hare]];
12     } while (tortoise != hare);
13
14
15     tortoise = arr[0];
16     while (tortoise != hare) {
17         tortoise = arr[tortoise];
18         hare = arr[hare];
19     }
20
21     return tortoise;
22 }
23
24 int main() {
25     int n;
26
27     if (scanf("%d", &n) != 1 || n < 2) {
28         return 0;
29     }
30
31     int* arr = (int*)malloc(n * sizeof(int));
32     if (arr == NULL) {
33         return 1;
34     }
35
36     for (int i = 0; i < n; i++) {
37         if (scanf("%d", &arr[i]) != 1) {
38             free(arr);
39             return 1;
40         }
41     }
42
43     printf("%d\n", findDuplicate(arr, n));
44
45     free(arr);
46

```

```
46 |  
47 |     return 0;  
48 | }  
49 |
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on	Thursday, 24 September 2026, 10:31 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:36 AM
Time taken	4 mins 16 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      if (scanf("%d", &n) != 1) return 0;
6
7      int arr[n];
8      for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
9
10     int tortoise = arr[0], hare = arr[0];
11     do {
12         tortoise = arr[tortoise];
13         hare = arr[arr[hare]];
14     } while (tortoise != hare);
15
16     tortoise = arr[0];
17     while (tortoise != hare) {
18         tortoise = arr[tortoise];
19         hare = arr[hare];
20     }
21
22     printf("%d\n", tortoise);
23     return 0;
24 }
25

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on Thursday, 24 September 2026, 10:37 AM**State** Finished**Completed on** Thursday, 24 September 2026, 10:42 AM**Time taken** 4 mins 38 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

```
1
3 10 17 57
6 2 7 10 15 57 246
```

Output:

```
10 57
```

Input:

```
1
6 1 2 3 4 5 6
2 1 6
```

Output:

```
1 6
```

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int T;
5     scanf("%d", &T);
6
7     while (T--) {
8         int n1, n2;
9         scanf("%d", &n1);
10
11         int a[n1];
12         for (int i = 0; i < n1; i++)
13             scanf("%d", &a[i]);
14
15         // ...
```

```

15     scanf("%d", &n2);
16
17     int b[n2];
18     for (int i = 0; i < n2; i++)
19         scanf("%d", &b[i]);
20
21     int i = 0, j = 0;
22
23     while (i < n1 && j < n2) {
24         if (a[i] == b[j]) {
25             printf("%d ", a[i]);
26             i++;
27             j++;
28         }
29         else if (a[i] < b[j]) {
30             i++;
31         }
32         else {
33             j++;
34         }
35     }
36
37     printf("\n");
38 }
39
40 return 0;
41 }
42
43

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on Thursday, 24 September 2026, 10:57 AM**State** Finished**Completed on** Thursday, 24 September 2026, 10:58 AM**Time taken** 37 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

```
1
3 10 17 57
6 2 7 10 15 57 246
```

Output:

```
10 57
```

Input:

```
1
6 1 2 3 4 5 6
2 1 6
```

Output:

```
1 6
```

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main() {
4     int T;
5     scanf("%d", &T);
6
7     while (T--) {
8         int n1, n2;
9
10        scanf("%d", &n1);
11
12        int a[n1];
13        for (int i = 0; i < n1; i++) {
14            scanf("%d", &a[i]);
15        }
```

```

15     }
16
17     scanf("%d", &n2);
18
19     int b[n2];
20     for (int i = 0; i < n2; i++) {
21         scanf("%d", &b[i]);
22     }
23
24     int i = 0, j = 0;
25     int first = 1;
26
27     while (i < n1 && j < n2) {
28         if (a[i] == b[j]) {
29             if (!first) {
30                 printf(" ");
31             }
32             printf("%d", a[i]);
33             first = 0;
34             i++;
35             j++;
36         }
37         else if (a[i] < b[j]) {
38             i++;
39         }
40         else {
41             j++;
42         }
43     }
44     printf("\n");
45 }
46
47
48 return 0;
49 }
50

```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on Thursday, 24 September 2026, 11:08 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:09 AM**Time taken** 1 min 10 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     int k;
14     scanf("%d", &k);
15
16     int i = 0, j = 1;
17
18     while (i < n && j < n) {
19         if (i == j) {
20             j++;
21         }
22         else if (a[j] - a[i] == k) {
23             printf("1");
24             return 0;
25         }
26         else if (a[j] - a[i] < k) {
27             j++;
28         }
29         else {
30             i++;
31         }
32     }

```

```
33 |
34 |     printf("0");
35 |
36 |     return 0;
37 | }
38 |
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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J2

Started on Thursday, 24 September 2026, 11:17 AM

State Finished

Completed on Thursday, 24 September 2026, 11:18 AM

Time taken 34 secs

Marks 1.00/1.00

Grade 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     int k;
14     scanf("%d", &k);
15
16     int i = 0;
17     int j = 1;
18
19     while (i < n && j < n) {
20         if (i == j) {
21             j++;
22         }
23         else if (a[j] - a[i] == k) {
24             printf("1");
25             return 0;
26         }
27         else if (a[j] - a[i] < k) {
28             j++;
29         }
30         else {
31             i++;
32         }

```

```
33     }
34
35     printf("0");
36
37     return 0;
38 }
39
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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